AIDS-1 Module 4

Explain various stages in the Data analytics Lifecycle. (10 marks)

The data analytics lifecycle consists of six sequential stages that guide analytics projects from conception to deployment.

1. Discovery: Define the problem, formulate initial hypotheses, and identify available data sources. Stakeholders assess project feasibility and success criteria.

2. Data Preparation: Extract, transform, and load (ETL) data from multiple sources. This stage includes data cleaning, handling missing values, outlier treatment, and converting data into a usable format.

3. Model Planning: Select appropriate analytical techniques such as regression, classification, clustering, or time series analysis. Data is sampled and partitioned into training and testing sets.

4. Model Building: Execute the planned models using tools like Python, R, or SAS. This stage involves iterative experimentation, parameter tuning, and comparing model performance metrics.

5. Results Communication: Interpret model outputs and translate findings into business recommendations. Visualizations, dashboards, and presentations are created for stakeholders.

6. Operationalization: Deploy the model into production environments. This includes monitoring model performance, establishing maintenance schedules, and documenting the entire process for future reference.

Each stage is iterative; insights from later stages may require revisiting earlier stages.

In detail, explain steps in the Data Science Project. (10 marks)

A data science project follows a structured methodology to ensure reliable and actionable outcomes.

Step 1: Problem Definition – Collaborate with stakeholders to understand business objectives, define success metrics (e.g., accuracy, ROI), and formulate the problem as a data science task (classification, regression, recommendation, etc.).

Step 2: Data Collection – Identify internal and external data sources: databases, APIs, web scraping, log files, or third-party datasets. Ensure data access rights and document provenance.

Step 3: Data Cleaning and Preprocessing – Handle missing values (imputation or deletion), remove duplicates, correct inconsistent formats, detect and treat outliers, and standardize or normalize numerical features.

Step 4: Exploratory Data Analysis (EDA) – Use summary statistics and visualizations (histograms, box plots, correlation matrices) to understand distributions, identify patterns, and uncover relationships between variables.

Step 5: Feature Engineering and Selection – Create new features from existing data (e.g., date parts, interaction terms), encode categorical variables, scale numerical features, and select the most relevant features using techniques like PCA or mutual information.

Step 6: Model Selection and Training – Split data into training (typically 60-80%), validation, and test sets. Train multiple candidate models (linear regression, random forest, neural networks) and tune hyperparameters using cross-validation.

Step 7: Model Evaluation – Assess performance using appropriate metrics (accuracy, precision, recall, F1-score, RMSE) on the test set. Compare against baseline models and business success criteria.

Step 8: Deployment and Monitoring – Package the model as an API or integrate into existing systems. Monitor drift in data distribution or prediction quality, and schedule retraining as needed.

Step 9: Documentation and Communication – Create technical documentation, stakeholder reports, and dashboards. Document assumptions, limitations, and maintenance procedures.

Differentiate between data scientists, big data professionals and data analysts. (10 marks)

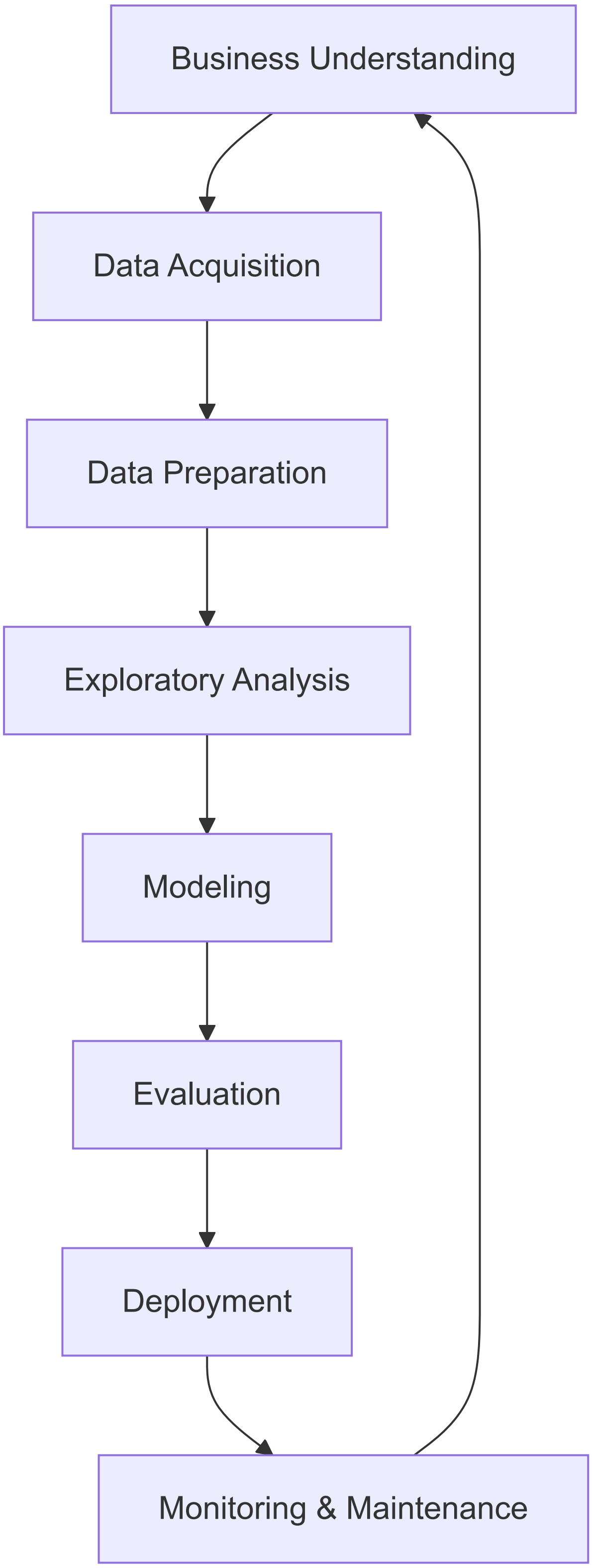
Aspect	Data Scientist	Big Data Professional	Data Analyst
Primary Focus	Predictive modeling, machine learning, and advanced analytics	Infrastructure, distributed computing, and data pipelines	Descriptive analytics, reporting, and business intelligence
Key Skills	Python/R, statistics, ML algorithms, SQL, experimentation	Hadoop, Spark, Kafka, cloud platforms (AWS/GCP/Azure), Scala/Java	SQL, Excel, Tableau/Power BI, data visualization, business acumen

Aspect	Data Scientist	Big Data Professional	Data Analyst
Typical Tasks	Building prediction models, A/B testing, feature engineering, algorithm optimization	Designing data architectures, ETL pipelines, managing clusters (HDFS, Spark), data streaming	Creating dashboards, writing SQL queries, generating reports, trend analysis
Data Volume	Works with moderate to large datasets (sampling common)	Specifically handles massive datasets (terabytes to petabytes)	Typically works with structured, medium-sized data
Output	Predictive models, actionable insights, ML APIs	Scalable data infrastructure, real-time processing systems	Static reports, interactive dashboards, KPI tracking
Programming	Python, R, SQL	Java, Scala, Python, Spark	SQL, basic Python/R

Example: A data analyst might report monthly sales trends. A data scientist builds a churn prediction model. A big data professional sets up a Spark cluster to process customer clickstream logs from millions of users.

Describe in detail the stages in the Data Science Lifecycle. (10 marks)

The data science lifecycle is an end-to-end framework for delivering data-driven solutions.



Stage 1: Business Understanding – Define objectives, success criteria (quantitative metrics), project constraints, and stakeholder requirements. Translate business problems into analytical tasks.

Stage 2: Data Acquisition – Identify relevant data sources (databases, APIs, files, external providers). Collect raw data while ensuring legality, privacy compliance, and representativeness.

Stage 3: Data Preparation – Clean data (missing values, duplicates, outliers), transform data types, merge datasets, and perform feature engineering. This typically consumes 60-

80% of project time.

Stage 4: Exploratory Data Analysis – Generate summary statistics, visualize distributions (histograms, box plots, scatter plots), detect anomalies, and test initial hypotheses. EDA guides modeling decisions.

Stage 5: Modeling – Select algorithms (regression, classification, clustering, etc.), split data (train/validation/test), train models, tune hyperparameters, and compare performance using cross-validation.

Stage 6: Evaluation – Assess model performance on unseen test data using appropriate metrics (accuracy, F1, RMSE). Validate against business success criteria and check for overfitting or bias.

Stage 7: Deployment – Integrate model into production environment (API, batch scoring, embedded system). Implement version control, logging, and rollback procedures.

Stage 8: Monitoring & Maintenance – Track model drift, data quality changes, and performance degradation. Schedule regular retraining and update documentation.

The lifecycle is iterative; insights from later stages frequently trigger revisiting earlier stages.

Compare and contrast Data Science, Business Analytics, and Big Data with examples. (10 marks)

Dimension	Data Science	Business Analytics	Big Data
Primary Goal	Extract insights and build predictive models from complex data	Support decision-making through descriptive and diagnostic analytics	Process and store massive, high-velocity datasets
Data Characteristics	Structured and unstructured (text, images, sensor data)	Primarily structured (tables, spreadsheets)	Volume, Velocity, Variety (3 Vs)
Techniques	Machine learning, deep learning, statistical modeling, experimentation	SQL, dashboards, regression, forecasting, OLAP	Distributed computing (Hadoop, Spark), NoSQL databases, stream processing
Time Orientation	Predictive (future) and prescriptive (what to do)	Descriptive (past) and diagnostic (why it happened)	Real-time or near-real-time processing
Output	ML models, recommendation	Reports, KPIs, dashboards, ad-hoc queries	Scalable data pipelines, data lakes, real-time

Dimension	Data Science	Business Analytics	Big Data
	engines, classification systems		analytics infrastructure
Typical Tools	Python, R, TensorFlow, PyTorch, Jupyter	SQL, Excel, Tableau, Power BI, SAS	Hadoop, Spark, Kafka, Hive, Cassandra
Example	Building a fraud detection model that learns from transaction patterns	Creating a monthly sales report with year-over-year comparison	Processing 1 million social media posts per second using Spark Streaming

Convergence: These fields overlap extensively. A big data infrastructure enables data science on massive scales. Business analytics often uses data science techniques for forecasting. The distinctions lie in primary focus: infrastructure (Big Data), prediction (Data Science), and reporting (Business Analytics).

What do you mean by data analytics? What are the different types of data analytics? (10 marks)

Data analytics is the systematic process of collecting, cleaning, transforming, and modeling data to discover meaningful patterns, draw conclusions, and support decision-making. It transforms raw data into actionable intelligence.

Four types of data analytics:

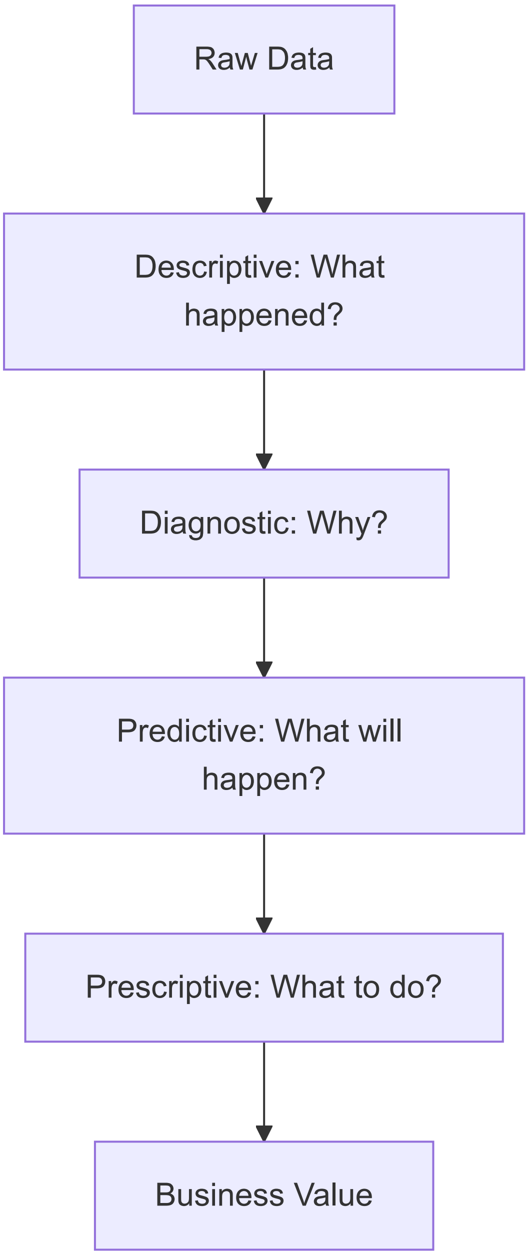
1. Descriptive Analytics – *What happened?*
Summarizes historical data using aggregates and visualizations. Examples: monthly sales reports, website traffic dashboards, student grade summaries. Tools: Excel, Tableau, Power BI.

2. Diagnostic Analytics – *Why did it happen?*
Identifies root causes and relationships through drill-down, data mining, and correlation analysis. Examples: analyzing why sales dropped in Q3, investigating customer churn drivers. Techniques: hypothesis testing, cohort analysis.

3. Predictive Analytics – *What is likely to happen?*
Uses statistical models and machine learning to forecast future outcomes. Examples: customer churn prediction, demand forecasting, loan default risk assessment. Techniques: regression, time series, classification.

4. Prescriptive Analytics – *What should we do about it?*
Recommends actions using optimization and simulation algorithms. Examples: dynamic

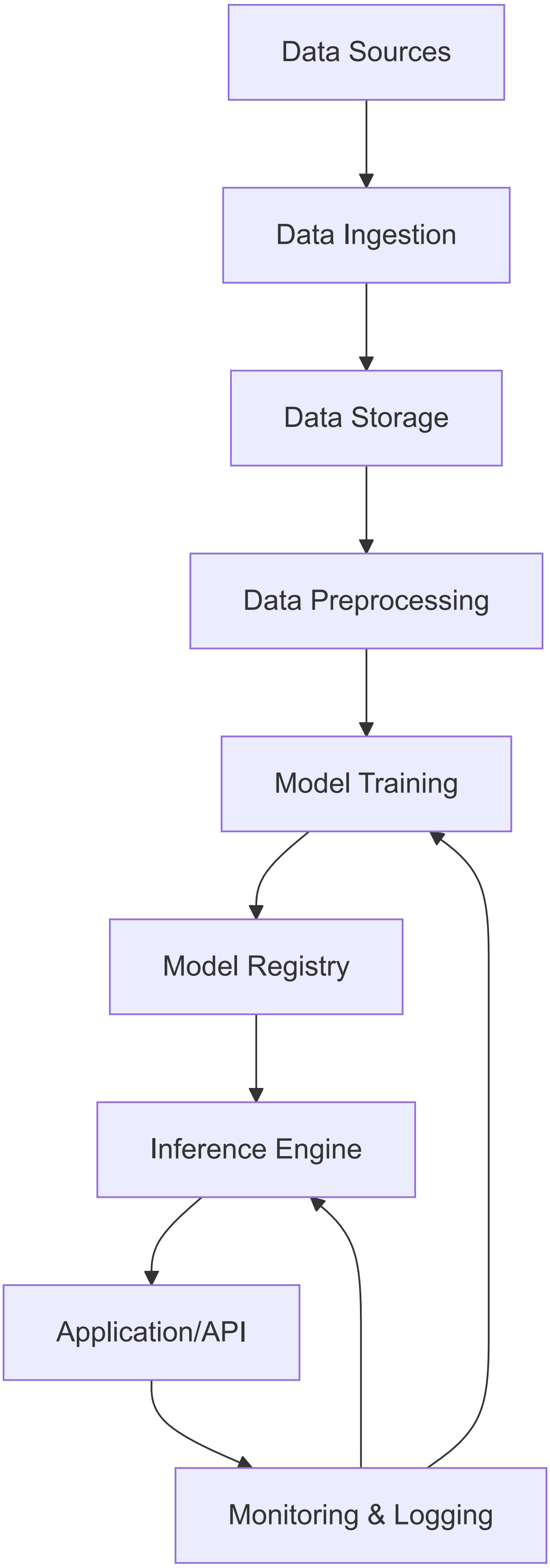
pricing, inventory optimization, treatment recommendations. Techniques: linear programming, reinforcement learning, decision trees.



Organizations typically progress from descriptive to prescriptive as analytical maturity increases.

Describe the architecture of ML application. Explain with a diagram. (10 marks)

Machine learning application architecture consists of interconnected components that handle data flow, model training, inference, and monitoring.



Component 1: Data Sources – Databases, APIs, IoT sensors, log files, cloud storage (S3, GCS), or streaming platforms (Kafka).

Component 2: Data Ingestion – Batch processing (Airflow, Luigi) or real-time streaming (Kafka, Kinesis). Handles data validation and initial formatting.

Component 3: Data Storage – Data lakes (raw data), data warehouses (processed data), or feature stores (curated features). Examples: Snowflake, BigQuery, Delta Lake.

Component 4: Data Preprocessing – Cleaning (missing values, outliers), transformation (scaling, encoding), feature engineering, and train/validation/test splitting.

Component 5: Model Training – Compute environment (GPU/CPU clusters) running ML frameworks (TensorFlow, PyTorch, Scikit-learn). Includes hyperparameter tuning and experiment tracking (MLflow, Weights & Biases).

Component 6: Model Registry – Version-controlled repository storing trained models, metadata, and performance metrics. Examples: MLflow Model Registry, Seldon.

Component 7: Inference Engine – Serves predictions via batch (scheduled) or real-time (REST API, gRPC) modes. Handles request batching and scaling.

Component 8: Application/API – End-user interface or external system consuming predictions. Manages authentication, rate limiting, and payload validation.

Component 9: Monitoring & Logging – Tracks data drift, concept drift, prediction latency, and model performance. Triggers retraining or alerts.

Differentiate between data scientists, big data professionals and data analysts. (10 marks)

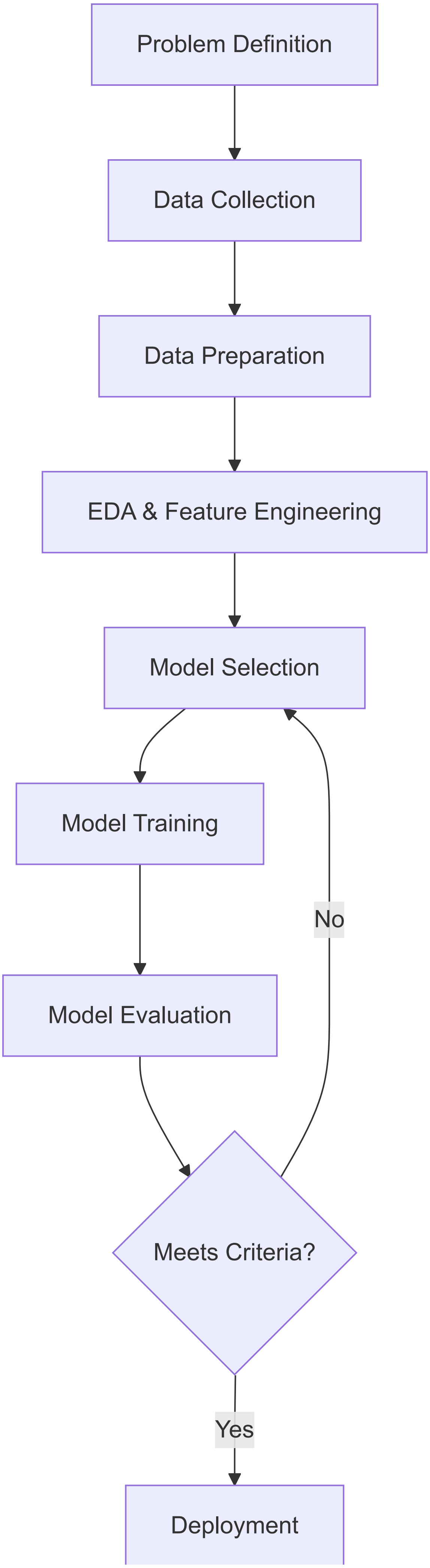
Aspect	Data Scientist	Big Data Professional	Data Analyst
Primary Role	Build predictive models and discover insights from complex data	Design, implement, and maintain distributed data processing systems	Query and visualize data to answer business questions
Core Techniques	Machine learning, statistical inference, A/B testing, deep learning	Distributed computing (MapReduce, Spark), stream processing, NoSQL databases	SQL, data visualization, descriptive statistics, reporting
Typical Outputs	Prediction APIs, classification models, recommendation engines, experiment results	Scalable ETL pipelines, data lakes, real-time processing infrastructure	Dashboards, periodic reports, ad-hoc query results, KPI tracking
Programming Languages	Python, R, SQL, sometimes Julia or	Java, Scala, Python, SQL, Bash	SQL, basic Python, occasionally R

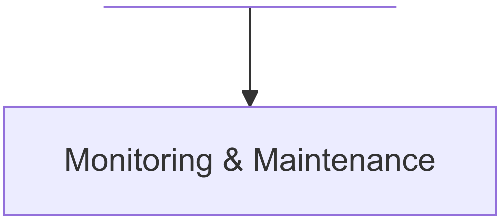
Aspect	Data Scientist	Big Data Professional	Data Analyst
	Scala		
Data Volume	Works with sampled or aggregated data (GB to TB)	Handles raw, massive datasets (TB to PB)	Works with curated, moderate-sized data (MB to GB)
Statistical Depth	Advanced (inferential statistics, probability, experimental design)	Minimal (basic descriptive statistics)	Intermediate (aggregations, trends, basic hypothesis testing)
Business Interaction	Moderate – translates business problems to technical tasks	Low – focuses on infrastructure reliability	High – directly supports business stakeholders
Example Project	Predict customer lifetime value using historical behavior	Set up a Spark cluster to process 500 GB of clickstream logs hourly	Create a monthly dashboard of sales by region and product category

Overlap note: In small organizations, one person may perform multiple roles. In large enterprises, these roles are distinct and collaborative.

Describe steps for developing ML applications with a labeled diagram. (10 marks)

Developing machine learning applications follows a systematic pipeline from problem identification to production deployment.


Problem Definition
Data Collection
Data Preparation
EDA & Feature Engineering
Model Selection
Model Training
Model Evaluation
Meets Criteria?
No
Yes
Deployment



Step 1: Problem Definition – Frame the business need as an ML task (classification, regression, clustering, recommendation). Define success metrics (accuracy, precision, recall, RMSE, business KPIs). Identify constraints (latency, interpretability, compute budget).

Step 2: Data Collection – Gather data from databases, APIs, web scraping, or sensors. Ensure sufficient volume, variety, and representativeness. Document data provenance and legal compliance.

Step 3: Data Preparation – Handle missing values (impute or delete), remove duplicates, correct data types, detect and treat outliers, normalize or standardize numerical features. Split into training (60%), validation (20%), and test (20%) sets.

Step 4: Exploratory Data Analysis & Feature Engineering – Visualize distributions, correlations, and patterns. Create new features from existing data (polynomial features, interaction terms, date components). Encode categorical variables (one-hot, label encoding). Select relevant features using statistical tests or feature importance.

Step 5: Model Selection – Choose candidate algorithms based on problem type, data size, interpretability needs, and computational constraints. Examples: linear regression for forecasting, random forest for classification, LSTM for time series.

Step 6: Model Training – Train models on training data using appropriate loss functions and optimizers. Tune hyperparameters using grid search, random search, or Bayesian optimization with cross-validation.

Step 7: Model Evaluation – Assess performance on the validation set using chosen metrics. Compare against baseline models. Check for overfitting (training vs validation performance gap), bias, and variance.

Step 8: Iteration – If criteria not met, revisit feature engineering, model selection, or hyperparameter tuning. Document all experiments.

Step 9: Deployment – Package the final model (pickle, ONNX, TensorFlow SavedModel). Create inference API using Flask, FastAPI, or TensorFlow Serving. Implement version control and rollback strategy.

Step 10: Monitoring & Maintenance – Track prediction latency, data drift, concept drift, and model accuracy in production. Establish retraining schedules (periodic or trigger-based). Log predictions for audit and improvement.